

Chapter 03 Homework — Rings

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Problem 03.1. Suppose that S is a set and R is a ring. Let R^S denote the set of all mappings from S to R . Show that R^S is a ring, under the operations defined by

$$(f + g)(s) = f(s) + g(s), \quad (fg)(s) = f(s)g(s).$$

Show that if S has more than one element then there exist non-zero elements f and g in R^S for which $fg = 0$.

Solution. Write $f_1, f_0 \in R^S$ for the functions

$$f_1(s) = 1_R, \quad f_0(s) = 0_R,$$

and clearly, by the specified function properties, we have all ring properties.

Prove $(f + g)h = fh + gh$. Let $s \in S$. Show $((f + g)h)(s) = (fh)(s) + (gh)(s)$:

$$((f + g)h)(s) = (f(s) + g(s))h(s) = f(s)h(s) + g(s)h(s) = (fh)(s) + (gh)(s).$$

If $|S| > 1$, there exist $s_1, s_2 \in S$ where $s_1 \neq s_2$. Let

$$f(s) = \begin{cases} 1_R & s = s_1 \\ 0_R & \text{else} \end{cases} \quad g(s) = \begin{cases} 1_R & s = s_2 \\ 0_R & \text{else} \end{cases}$$

Then $f(s)g(s) = 0$, so they are zero divisors.

Problem 03.2. Suppose that R is an integral domain, with field of fractions F . Show that the field of fractions of $R[x_1, \dots, x_n]$ can be identified naturally with $F(x_1, \dots, x_n)$.

Problem 03.3. *[TODO: statement not yet transcribed]*

Problem 03.8. Suppose that K is a field. If $f = a_0 + a_1x + \dots + a_nx^n \in K[x]$ and $k \in K$, let $(\Phi(f))(k) = a_0 + a_1k + \dots + a_nk^n$. Show that Φ is a ring homomorphism from $K[x]$ to K^K . Show that if K is finite then Φ is an epimorphism, but not a monomorphism. What happens if K is infinite?

Solution. Let $f = \sum_i a_i x^i$ and $g = \sum_i b_i x^i$ in $K[x]$.

Additive half. Let $k \in K$. Then

$$\begin{aligned} (\Phi(f + g))(k) &= \left(\Phi\left(\sum_i (a_i + b_i)x^i\right) \right)(k) = \sum_i (a_i + b_i)k^i = \sum_i a_i k^i + \sum_i b_i k^i \\ &= \left(\Phi\left(\sum_i a_i x^i\right) \right)(k) + \left(\Phi\left(\sum_i b_i x^i\right) \right)(k) = (\Phi(f))(k) + (\Phi(g))(k). \end{aligned}$$

As $k \in K$ was arbitrary, $\Phi(f + g) = \Phi(f) + \Phi(g)$ in K^K .

Multiplicative half. Let $k \in K$. Then

$$\begin{aligned} (\Phi(fg))(k) &= \left(\Phi \left(\sum_m \left(\sum_{i+j=m} a_i b_j \right) x^m \right) \right)(k) = \sum_m \left(\sum_{i+j=m} a_i b_j \right) k^m \\ &= \left(\sum_i a_i k^i \right) \left(\sum_j b_j k^j \right) = \left(\Phi \left(\sum_i a_i x^i \right) \right)(k) \cdot \left(\Phi \left(\sum_j b_j x^j \right) \right)(k) = (\Phi(f))(k) (\Phi(g))(k). \end{aligned}$$

Φ is an epimorphism when K is finite. Let K be a finite field and let $h \in K^K$ be arbitrary. We find $f \in K[x]$ with $(\Phi(f))(k) = h(k)$. Take

$$f(x) = \sum_k h(k) \delta_k(x).$$

Let $k_1 \in K$. Then

$$(\Phi(f))(k_1) = \left(\Phi \left(\sum_k h(k) \delta_k(x) \right) \right)(k_1) = \sum_k h(k) \delta_k(k_1) = h(k_1) \delta_{k_1}(k_1) = h(k_1) \cdot 1 = h(k_1).$$

So Φ is an epimorphism.

Φ is not a monomorphism. Take $K = \mathbb{F}_2$, $f = x^2 + x$ and $g = 0$. Then $f \neq g$ in $\mathbb{F}_2[x]$, but

$$(\Phi(f))(0) = 0 = (\Phi(g))(0), \quad (\Phi(f))(1) = 1 + 1 = 0 = (\Phi(g))(1),$$

so $\Phi(f) = \Phi(g)$ in $\mathbb{F}_2^{\mathbb{F}_2}$ and $\ker \Phi \neq \{0\}$.

For an arbitrary finite field K , take

$$f(x) = \prod_{k \in K} \delta_k(x),$$

a non-zero element of $\ker \Phi$.

Problem 03.10. What are the units in $R[x]$, where R is an integral domain? What are the units in $\mathbb{Z}_4[x]$?

Solution. The units in $R[x]$, for R an integral domain, are

$$\{a \in R \subseteq R[x] : a^{-1} \text{ exists}\},$$

the existence of a^{-1} being what is not guaranteed in an integral domain.

They must be constants. For in an integral domain R , with $f, g \in R[x]$,

$$\deg(fg) = \deg(f) + \deg(g),$$

since no non-zero coefficients multiply to zero. So $\deg(f) = \deg(g) = 0$.

In $\mathbb{Z}_4[x]$ the units are 1 and 3, since $3 \cdot 3 = 9 \equiv 1$; and also, taking

$$f(x) = 2x + 3, \quad g(x) = 2x + 3,$$

we have $fg = 4x^2 + 12x + 9 = 1$.

So in $\mathbb{Z}_4[x]$ the units are those f whose constant term is a unit and all of whose other coefficients are zero divisors: constant term in $\{1, 3\}$, every other coefficient in $\{0, 2\}$.

Problem 03.11. Show that an element a of an integral domain R is prime if and only if $R/(a)$ is an integral domain.

Solution. ($\Rightarrow$) Suppose $a \in R$ is prime. Show $R/(a)$ is an integral domain.

Consider

$$(r_1 + (a))(r_2 + (a)) = r_1r_2 + (a).$$

Take the factors non-zero, so $a \nmid r_1$ and $a \nmid r_2$ — else we would have zero divisors. Then $r_1r_2 + (a)$ is non-zero, as $a \nmid r_1r_2$ since a is prime. So $R/(a)$ has no zero divisors.

Further, $1 + (a) \neq 0 + (a)$, because $a \nmid 1$ (a is not a unit).

($\Leftarrow$) Suppose $R/(a)$ is an integral domain. Show $a \in R$ is prime.

Suppose a is not prime. So there exist $b, c \in R$ such that $a \mid bc$, but $a \nmid b$ and $a \nmid c$. Note $b + (a)$ and $c + (a)$ lie in $R/(a)$, both non-zero since $a \nmid b$, $a \nmid c$. And

$$(b + (a))(c + (a)) = bc + (a) = (a)$$

since $a \mid bc$. So $R/(a)$ has zero divisors and is not an integral domain.

Problem 03.12. *[TODO: statement not yet transcribed]*

Problem 03.14. *[TODO: statement not yet transcribed]*

Problem 03.15. *[TODO: statement not yet transcribed]*

Problem 03.17. *[TODO: statement not yet transcribed]*

Problem 03.25. *[TODO: statement not yet transcribed]*

Problem 03.26. *[TODO: statement not yet transcribed]*